

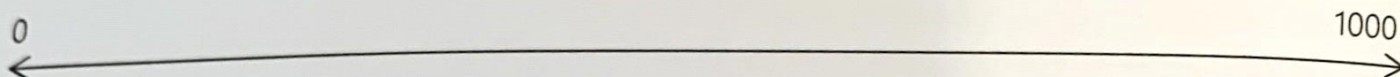
1. Relative size of numbers to 1000

RELATIVE SIZE

Relative size underpins concepts such as fractions, percent, converting measurements and even probability. It is a fundamental idea that we will use to build many other concepts.

Experiment

Copy the number line beneath on a large piece of paper. Where would these numbers fit?
10, 100, 200



Does your first idea look right? Try adding in all the other hundreds and consider if the spacing is reasonable. It is not unusual to try 10 different lines before the spacing works.

This Number Line strategy will be used frequently in later lessons and chapters.

Explore

Stimulus questions

- How many hundreds make up 1000?
- How many tens make up 100?
- Where would 110 go? 120? 130?
- How about 210? 220? 230?

Patterns and connections

What do you think should go in the exact middle of the line? How come? Redraw your line below your original line with this in mind (only if you need to change your mind).

Evaluate

What key numbers or position points can be used to help you figure out positions on a number line from 0 to 1000? If you changed your mind, what did you change your mind about?

Explain

There is only one correct position for each number.

- How do you know that your numbers are in the right place?
- What numbers or positions would you start with next time?

Extend

- Use a strip of paper one metre long. Create your number line on this strip, making sure that the scale is correct by folding the paper into the correct size partitions. You will need this for later lessons.

Work with a friend on at least two of the tasks below:

- What would happen if your number line scale changed to 0-100, 0-10 or 0-1 on the original line? What key numbers or position points would be used to help position other numbers?
- What would change if your number line scale was to 0-2000? What key numbers or position points would be helpful?
- What if your number line showed a bank balance where you could borrow up to \$1000? How could you apply similar thinking to show negative balances?

Apply relative size in real life situations

Set A: Length

Use the one metre long **Number Line** that you made to measure the length and width of the following items. Notice that 0-1000 is the distance in millimetres. A metre-long line from 0-100 would show centimetres.

- 1. Your desk
- 2. Your classroom
- 3. Your pen or pencil

Set B: Convert units of length

Use your one metre long number line to consider the connections between the following amounts. Complete the following table to show conversions for units. You don't need to follow any rules however you should try to find connections and to explain any that you notice.

Metres	Centimetres	Millimetres	What connections can you see?
	7cm		
		7mm	
7m			
	75cm		
		75mm	

The strategy shown below is called using a **Place Value Chart**. It will help you with conversions. **This strategy will be used frequently in later chapters.**

Thousands (of ones)	Hundreds (of ones)	Tens (of ones)	Ones. (of ones)	Tenths (of ones)	Hundredths (of ones)	Thousandths (of ones)
kilo			Base unit (m)		centi	milli

Set C: Integers, negative numbers and money

A **Number Line** visually representing a bank balance between -100 to +100, with zero in the middle, is a great way to visualise negative numbers. The negative end represents owing money, such as spending with a credit card. Sketch the line and use it to work out the answers to the following questions.

- 1. I have \$80 in the bank.
- 2. I use a credit card and spend \$100. Draw an arrow to show what happens on the line and mark the end position.
- 3. I spend another \$20. Draw an arrow to show what happens and mark the end position.
- 4. I deposit (add in) \$50 into my bank account. Draw an arrow to show what happens and mark the end position. **What is my bank balance?**

2. Efficient addition strategies

PARTITIONING

Many addition strategies that we were taught have been inherited from times before technology was around to help with complex situations. This lesson is designed to teach more efficient strategies for simple addition situations, starting with the largest place. These strategies are also great for approximation and easy to carry out mentally. They will be used in other chapters.

Explore

Vertical format, largest place first

Instead of adding from the right, add from the left and write a number on each line below.

Use the worked example to understand how the strategy works, then try the questions set.

1. $347 + 492$
2. $37 + 94$
3. $2508 + 3892$
4. $347 + 492 + 156$
5. $47 + 492$
6. $237 + 94$
7. $508 + 3892$
8. $347 + 92 + 156$

Worked example

	2	7	3
+	3	6	5
	5	0	0
	1	3	0
			8
	6	3	8

Horizontal Format Flexible Strategy

This strategy is very similar to the first but involves a different layout. It is particularly useful as you can also use it for negative numbers, fractions and decimals. Use the worked example to understand how the strategy works, then try the questions set. They use the same numbers as Set B above. **This Horizontal Format strategy will be used in later lessons.**

9. $47 + 492$
10. $237 + 94$
11. $508 + 3892$
12. $347 + 92 + 156$

Worked example

$$273 + 365$$

Let's think hundreds, tens, ones

$$= 500 + 130 + 8$$

$$= 638$$

Evaluate

1. What did you like about these strategies? Which did you prefer? How do you think starting with the largest numbers might be useful for mental calculations rather than written calculations?
2. Is there a way that you could use just your first calculation to come up with a reasonable range for possible answers without calculating the whole lot? Think: "It will be at least... and not more than...". **This strategy is called Leading-Digit Approximation and will be used in later lessons.**

3. Relative size of negative numbers

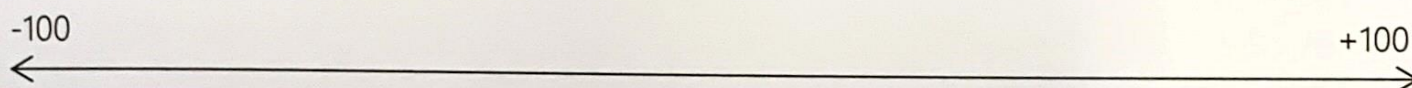
RELATIVE SIZE

Negative numbers are very important in finance. We often use credit cards and loans, so need to understand what happens with loan balances when we deposit (add money in) or withdraw/spend (take money out). This concept was introduced in Set C from Relative size to 100.

Experiment

Apply the **Number Line Strategy**. Where would the following numbers fit on the scale below? Try using a very large piece of paper.

-50, 25, 0, -75, 50, 75



Explore

Make strips of paper that are 50, 25 and 75 long when compared to your number line. Using those strips of paper and your number line, write as many number sentences as you can using just those numbers.

Stimulus questions

Write equations for these situations and answer them:

- Start at 75, end at 25: what happened?
- Start at 25, end at 75: what happened?
- Start at 25, end at -50: what happened?
- Start at -50, end at 25: what happened?

Patterns and connections

Find the connections between the four stimulus questions above.

Extend

Work with a friend to draw a line from -2 to +2.

Place the following decimal numbers on the line: 0.5, 1.5, -0.5, -1.5, 0.25, -0.25, 1.25, -1.25

Write a hard question that someone who had good understanding of negative numbers could answer. What would you want to see in their answer?

Evaluate

Use your strips of paper, the number sentences you came up with in the Explore section, and the patterns you found to evaluate the following:

- How is 75-25 similar to 25-75? Why?
- Would this same pattern apply to other numbers? Try 8-3 and 3-8: what happens? Try 17-3 and 3-17: what happens?
- Would this pattern always be true or only sometimes?

Explain

If this pattern is always true, how could you use it to subtract larger numbers from smaller numbers?

4. Efficient subtraction strategies

🔗 PARTITIONING 🔗 RELATIVE SIZE

Understanding negative numbers means that we can subtract using much more efficient methods. Instead of starting with the ones, the following methods start with the largest numbers. These methods are simpler to perform mentally and make approximations easier as well.

Strategy one: start with the largest place, vertical format

Instead of subtracting from the right, start from the left and write a number on each line below. Use the worked example to understand how the strategy works, then try the questions set.

Set A:

1. $347 - 192$
2. $94 - 39$
3. $2508 - 1892$
4. $2001 - 901$

Set B:

5. $407 - 49$
6. $230 - 94$
7. $49 - 407$
8. $94 - 230$

Worked example

$$\begin{array}{r} 3 \quad 6 \quad 2 \\ - 1 \quad 7 \quad 5 \\ \hline 2 \quad 0 \quad 0 \\ - \quad 1 \quad 0 \\ - \quad \quad 3 \\ \hline 1 \quad 8 \quad 7 \end{array}$$

Strategy two: Horizontal Format

Apply the Horizontal Format strategy from lesson 2. Use the worked example to understand how the strategy works, then try the questions set. They use the same numbers as Set B above.

This strategy will be used in later lessons.

Set C:

9. $407 - 49$
10. $230 - 94$
11. $49 - 407$
12. $94 - 230$

Worked example

$$\begin{aligned} 362 - 175 \\ = 200 - 10 - 3 \\ = 187 \end{aligned}$$

You could also use a **Number Line** strategy and jump along the line to calculate the answers.

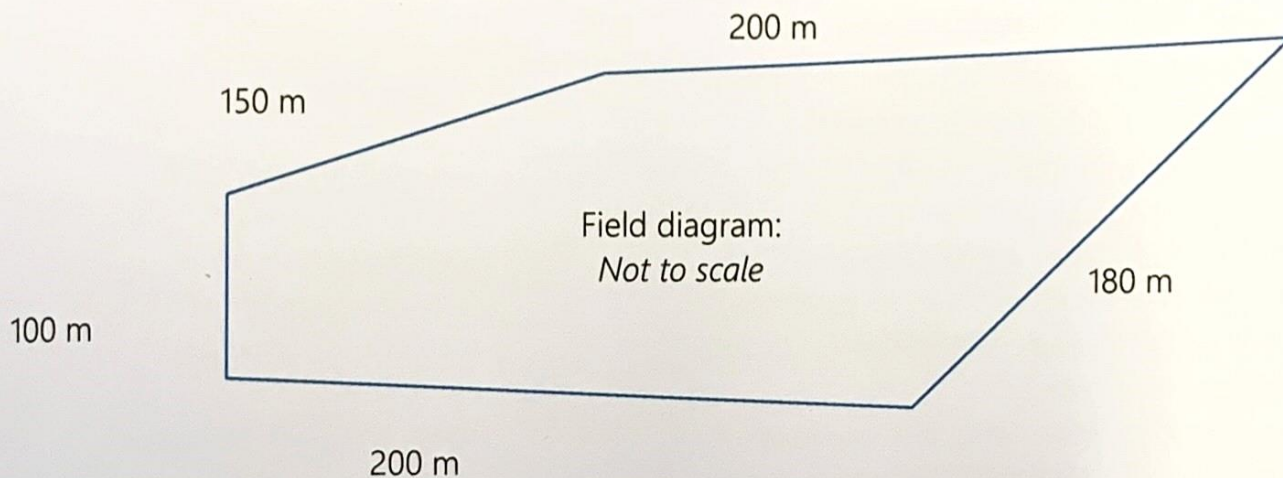
Reflect

1. What did you like about these strategies? Which did you prefer?
2. How do you think starting with the largest numbers might be useful for mental calculations rather than written calculations?
3. Look at your answers for each set and the first calculation that you did. Is there a way that you could use just your first calculation to come up with a reasonable estimate without calculating the whole lot? Think: "It will be at least... and not more than...". This is called "leading-digit approximation".

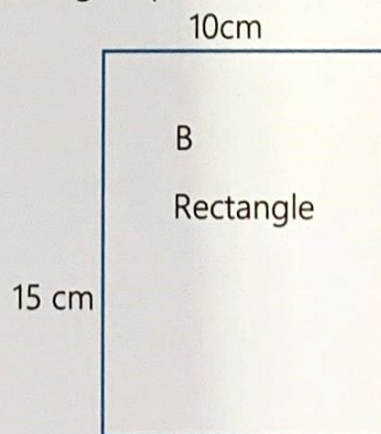
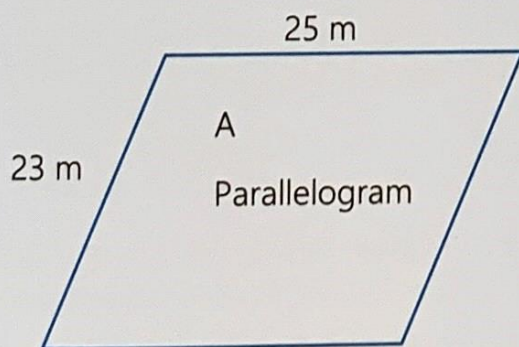
5. Application of addition and relative size to perimeter

 PARTITIONING  RELATIVE SIZE

Perimeter is one of the most common calculations used in building, renovating and allocating land. You don't need a formula, but you do need to apply your understanding of what we have been learning in this chapter to work it out. Use the following diagram to deduce what perimeter means and how to figure it out. Think about using your Number Line strategy to help.

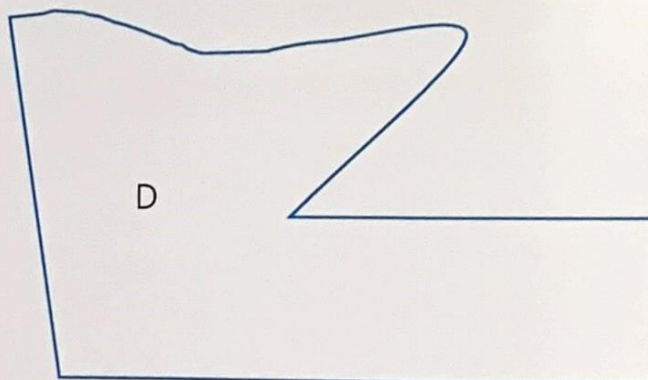
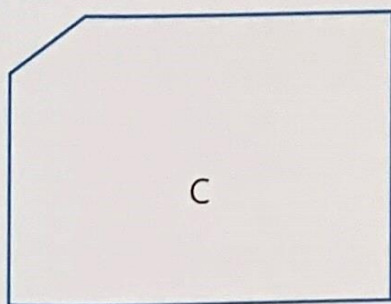


1. The perimeter of the field in the diagram above is 830m. How do we calculate perimeter?
2. Use the same method to work out the perimeter for the following shapes:



Set A: Use your **Number Line** metre strip to work out perimeter of the following

3. Your desk
4. Your maths book
5. A square that has one side of 7cm
6. A rectangle with a length of 12cm and a width of 8cm
7. These shapes:



6. Relative size with decimal numbers

RELATIVE SIZE PROPORTIONAL REASONING

*Understanding relative size with length helps us to unpack decimal numbers. The decimal point in a number only tells us where the ones are. Discuss the following stimulus questions, then use what you unpack to experiment with the problem below. This is the **Of A Dollar** strategy.*

Discussion questions

1. In our number system we work on a base number. This number is used when we have to move from ones to tens, or from tens to hundreds, or from hundreds to thousands because we have too many to fit in that place. What number is this?
2. Why do you think we chose 10 to be the basis for our number system? Hint: it has to do with our bodies.
3. Each time we move one digit to the left in our base ten system, the number is ten times larger. For example, 100 is ten times larger than 10. Likewise, each time we move one digit to the right, the number is ten times smaller. For example, 50 is ten times smaller than 500.

The same is true with decimal numbers:

- 0.1 is ten times bigger than 0.01
- 0.002 is ten times smaller than 0.02

RELATIVE SIZE

PROPORTIONAL REASONING

"Deci" is Latin for "tenth".

Decimal numbers are therefore "base ten". Money is also "base ten", so we can use it to understand converting fractions to decimals.

What is half Of A Dollar?

Write that in dollars and cents. That is how we write $\frac{1}{2}$ as a decimal number.

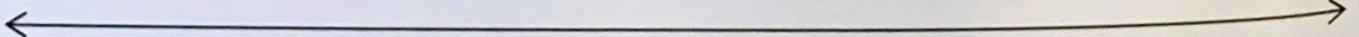
What is one quarter Of A Dollar?

Write that in dollars and cents. That is how we write $\frac{1}{4}$ as a decimal number.

Experiment

Where would the following numbers fit on a scale of 0 to 1? Draw the line below on a large piece of paper. Use the strategy **Of A Dollar** to convert the following fractions into decimals and write them on the line: one tenth, two tenths, one half, one quarter, one fifth, two fifths

0



Explore

Does your first idea look right? Try adding in all the other tenths and consider if the spacing is reasonable.

- Where would 50 hundredths fit?
- What do you notice about its position?
- What other decimal numbers and fractions are at the same position?

Evaluate and Explain

What key numbers or position points can be used to help you figure out positions on a number line from 0 to 1?

With whole numbers adding a zero to the right shifts all digits to the left, making the number ten times bigger. Why doesn't adding a zero to the right of a decimal number do the same thing and make it ten times bigger?

7. Apply "Relative Size" with decimals to estimation

RELATIVE SIZE PROPORTIONAL REASONING

When we understand the relative size of decimal numbers, it is easier to estimate and to figure out what a reasonable calculation would be for lengths. Your **Number Line** strategy will help.

Set A: Estimation and approximation

Use a strip of paper that is one metre long as a line from 0 to 1. Estimate where the following decimal numbers would be. Consider what decimal number is close to the amount you are given and what fraction is roughly the same. Mark them on your line and explain how you worked out where each should be placed.

- | | | |
|----------|-----------|------------------|
| 1. 0.496 | Close to: | Nearby fraction: |
| 2. 0.102 | Close to: | Nearby fraction: |
| 3. 0.253 | Close to: | Nearby fraction: |
| 4. 0.989 | Close to: | Nearby fraction: |
| 5. 0.81 | Close to: | Nearby fraction: |

Set B: Ordering and comparing lengths

Compare the following sets of measurements. Use the box on the right to help you to convert them into metres so that you can compare them.

Rewrite each set in ascending order (smallest to largest), including explaining if any numbers are equal.

6. 0.5m, 0.50m, 0.05m
7. 49mm, 4.9m, 49cm
8. 2.5cm, 250mm, 0.0025m
9. 0.5m, 5cm, 0.50cm

Set C: Rounding decimal numbers

In Set A above you considered what decimal number was close to the one you were given. This is an informal way of rounding. When we are rounding numbers, we consider which number on either side is closest to our amount. For example, if rounding 0.463 to the nearest one we would consider if it was closer to 0 or 1; if rounding it to the nearest tenth we would consider if it was closer to 0.4 or 0.5; if rounding it to the nearest hundredth we would consider if it was closer to 0.46 or 0.47. Complete the table to round the following numbers.

Original number	Nearest one	Nearest tenth	Nearest hundredth
0.463			
0.555			
0.494			
0.209			

RELATIVE SIZE

When converting between measurements it is helpful to think about whether you need more or less.

For example, consider 1m. There are *more* cm in a m, because cm are smaller so more fit into the same space. That means that after converting from metres into to centimetres, you should have a *larger* number.

m to cm: *you should have more*
 $1\text{m} = 1 \text{ (sets of 100) } = 100\text{cm}$
 $2\text{m} = 2 \text{ (sets of 100) } = 200\text{cm}$

cm to m: *you should have less*
 $500\text{cm} = 500 \text{ (split into sets of 100) } = 5\text{m}$
 $250\text{cm} = 250 \text{ (split into sets of 100) } = 2.5\text{m}$

8. Connecting fractions, decimals and percentage

🔗 RELATIVE SIZE 🔗 PROPORTIONAL REASONING

Fractions, decimal numbers and percentages are different ways of representing the same amount. When we looked at Relative Size of decimal numbers, we used the strategy **Of A Dollar** to help us turn fractions into decimals without requiring a formal process. In this task we will explore further connections between money, measurement, decimals, percentage and fractions to help solidify connections and make converting between them easy.

Explore

- How many **cents** are in each dollar?
- How many **centimetres** in each metre?
- How many years in each **century**?
- What number do you think the prefix **cent** represents?
- What do you think **percent** means?
- What **percent** is equal to one whole?

Find the pattern:

Consider half **Of A Dollar**:

How many cents in half a dollar?

Write the number of cents, and also write the amount in dollars.

- Write a half as a decimal number.
- What is half written as a percent?
- What pattern can you find?

Consider one quarter **Of A Dollar**:

How many cents in one quarter of a dollar?

Write the number of cents, and also write the amount in dollars.

- Write a quarter as a decimal number.
- What do you think one quarter might be when written as a percentage?
- What pattern can you find?

Consider one fifth **Of A Dollar**:

Write the number of cents, and also write the amount in dollars.

- Write a fifth as a decimal number.
- What do you think one fifth might be when written as a percentage?
- What pattern can you find?

Extend and Experiment

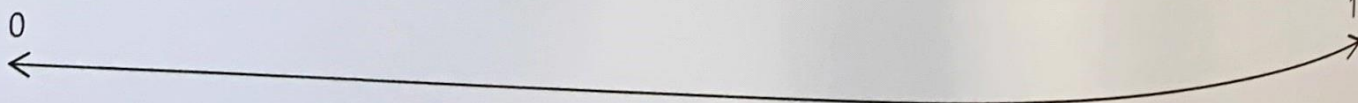
One particular operation will help you to convert fractions into decimals and therefore, into percentages. Use what you have worked out in the exploration section above to experiment with a calculator and find out which operation to use to turn your fractions into decimals.

For example, to turn $\frac{1}{2}$ into 0.5 you need to press these keys: $1 \div 2 =$

Work out what operation goes on the line.

Use either the **Of A Dollar** idea or the operation you found above to convert the following fractions into both decimals and percentages, then place them on a **Number Line**:

$\frac{1}{5}$ $\frac{1}{4}$ $\frac{1}{3}$ $\frac{2}{5}$ $\frac{2}{4}$ $\frac{2}{3}$ $\frac{3}{5}$ $\frac{3}{4}$



9. Apply fractions, decimals and percentage in the real world

 RELATIVE SIZE

 PROPORTIONAL REASONING

Work with a partner to apply your understanding of converting between the fractions, decimals and percentages to the following problem sets. Try to work out each answer mentally first, then use a calculator to check that you are on the right track.

Set A: Length of fabric

Fabric is purchased in centimetres. Convert each of the following amounts into centimetres.

- Two and a half metres ($2\frac{1}{2}$ as a mixed number)
- One and a quarter metres ($1\frac{1}{4}$ as a mixed number)
- Window frame measures 2750mm: write it in cm for fabric, in metres so that we can easily picture it, and as a mixed number for simple comparison of size.

Set B: Discounts

Discounts are often expressed as percentages. It is important to be able to think about how big each discount is. Express each of the discounts below as a fraction. The first is done for you as an example. **This strategy is called Partition It and you will use it frequently.**

Discount percentage	Fraction (Of A Dollar strategy)	Partition It: to work out what is left (you pay):
25% off 25c per dollar off	$\frac{1}{4}$ off	75% of the original price, or $\frac{3}{4}$ of the original price
20% off		
10% off		
		60% of the original price, or $\frac{6}{10}$ of the original price
	$\frac{4}{5}$ off	

Set C: Chance of events occurring

Chance or probability is expressed as a likelihood between 0 (will never happen) and 1 (will always happen). Work out the fraction, decimal and percentage for each of the following situations and place it on the number line from last lesson. Use a **Number Line** if it helps.

- Flipping a coin and having it land on heads
- Choosing a spade from a deck of cards
- Rolling a 3 on a die
- The sun rising tomorrow
- The sun rising again today
- Choosing the correct number between 1 and 10